

Multiple Choice

1. **Answer:** D

Options: A) $5.5 \times 10^{-13}g$; B) $2.5 \times 10^{-13}g$; C) $2.0 \times 10^{-12}g$; D) $4.7 \times 10^{-13}g$; E) $8.6 \times 10^{-13}g$

Note: Correct option : D - $4.7 \times 10^{-13}g$

1. **Answer:** A

Options: A) 1750.0; B) 1500.0; C) 2200.0; D) 2100.0; E) 1400.0

Note: Correct option: A - 1750.0

2. **Answer:** C

Options: A) two-thirds; B) twice; C) the same; D) half; E) four times higher

Note: Correct option: C - the same

3. **Answer:** A

Options: A) Virgo; B) Perseus; C) Ursa Major; D) Sagittarius; E) Aquila

Note: Correct option: A - Virgo

4. **Answer:** A

Options: A) 48 ft per sec; B) 64 ft per sec; C) 30 ft per sec; D) 72 ft per sec; E) 16 ft per sec

Note: Correct option: A - 48 ft per sec

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '25%'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'halve.'

Long Form Response

11. **Answer:** (a) circular, (b) rectangular. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 4.2 cm, 5.9. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key